CBN1 — CBN1 TASK 1: BIG DATA ARCHITECTURE PROPOSAL

DATA ANALYTICS AT SCALE — D609

PRFA — CBN1

Preparation | **Task Overview** | Submissions | Evaluation Report

COMPETENCIES

4169.1.1 : Applies Tools to Support Data Analysis at Scale

The learner explains the advantage of using a big data analytics solution.

4169.1.2 : Proposes a Large-Scale Data Analytics Solution

The learner proposes a large-scale data analytics solution to meet a business need.

INTRODUCTION

In this task, you will read and review the details of the STEDI Human Balance Analytics case study in the Udacity platform, then develop a proposal that simulates a big data analytics proposal to appropriate stakeholders. Due to contractual limitations, the organization is unable to build any big data analytics on AWS. They have an existing business relationship with Microsoft and are required to use Azure services to build big data analytics systems. Therefore, your proposed solution must use Microsoft platforms and Azure services to perform the required analyses.

The proposal will require the following components:

- an executive summary detailing the purpose and expected outcomes
- a recap of the identified business problem and its significance
- organizational needs assessment with a focus on how big data analytics can address these needs
- detailed data architecture plan, specifying databases and processing systems
- data visualization models demonstrating clarity and stakeholder understanding
- reasons behind architectural and technological choices
- summary of future enhancements or iterations
- a short implementation plan outlining proposed steps

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker. Students are strongly encouraged to wait for the similarity report to generate after uploading their work and then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See Understanding Similarity Reports for more information.

Help

Grammarly Note:
Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged to review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See Use Grammarly for Education Effectively for more information.

Microsoft Files Note:
Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see Computer System and Technology Requirements.

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

Part 1: Big Data Architecture Proposal

In this task you will create a proposal based on the STEDI Human Balance Analytics project that includes the following:

Executive Summary

A. Complete the executive summary by doing the following:
  1. Summarize the purpose of the proposal.
  2. Describe the anticipated outcomes, emphasizing organizational benefits.

Business Problem Recap

B. Summarize the identified business problem, highlighting the significance for the organization.

Needs Assessment

C. Complete the needs assessment by doing the following:
  1. Explain how big data analytics can address the identified organizational need.
  2. Describe data analysis methods specific to large datasets.

Solution Design

D. Design a solution by doing the following:
  1. Identify databases and large-scale processing systems needed to support the proposed analytics solution.
  2. Recommend large-scale data models to meet the needs of the business solution.
  3. Recommend a large-scale data analytics solution that integrates the databases and processing systems identified in D1 with querying, visualization, and reporting functionality.
  4. Describe any required event orchestration, security, or other ancillary services required as part of the proposed analytics solution.

Justification of Choices

E. Justify your choices by doing the following:
  1. Explain the reasoning behind each architectural and technological choice made during the design phase.
  2. Explain how the architectural and technological choices align with the business problem and organizational needs.

Future Enhancements

F. Describe potential future enhancements or iterations to further improve the big data architecture.

Implementation Plan

G. Provide a brief overview of the proposed steps for implementing the designed solution.

H. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

I. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ( )
File size limit: 200 MB
File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A1:PROPOSAL PURPOSE

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A summary of the purpose of the proposal is not provided.	The summary of the proposal's purpose is not aligned with the case study's details or contains factual or logical errors.	The summary of the proposal's purpose is aligned with the case study's details and contains no factual or logical errors.

A2:OUTCOMES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A description of the anticipated outcomes is not provided.	The description of the anticipated outcomes is not aligned with the case study's details or contains factual or logical errors.	The description of the anticipated outcomes is aligned with the case study's details and contains no factual or logical errors.

B:BUSINESS PROBLEM

NOT EVIDENT
A summary of the business problem is not provided.

APPROACHING COMPETENCE
The summary of the business problem is not aligned with the case study's details or contains factual or logical errors.

COMPETENT
The summary of the business problem is aligned with the case study's details and contains no factual or logical errors.

C1:ORGANIZATIONAL NEED

NOT EVIDENT
An explanation of how big data analytics can address the identified organizational need is not provided.

APPROACHING COMPETENCE
The explanation of how big data analytics can address the identified organizational need is not aligned with the case study's details or contains factual or logical errors.

COMPETENT
The explanation of how big data analytics can address the identified organizational need is aligned with the case study's details and contains no factual or logical errors.

C2:DATA ANALYSIS METHODS

NOT EVIDENT
A description of the data analysis methods specific to large datasets is not provided.

APPROACHING COMPETENCE
The description of the data analysis methods specific to large datasets is not aligned with the case study's details or contains factual or logical errors.

COMPETENT
The description of the data analysis methods specific to large datasets is aligned with the case study's details and contains no factual or logical errors.

D1:DATABASES AND PROCESSING SYSTEMS

NOT EVIDENT
An identification of databases and large-scale processing systems needed to support the proposed analytics solution is not provided.

APPROACHING COMPETENCE
The response fails to identify databases and large-scale processing systems, does not align with the case study's details, or contains factual or logical errors.

COMPETENT
The response identifies databases and large-scale processing systems needed to support the proposed analytics solution, aligns with the case study's details, and contains no factual or logical errors.

D2:DATA MODELS

NOT EVIDENT

APPROACHING COMPETENCE

COMPETENT

A recommendation for large-scale data models to meet the needs of the business solution is not provided.	The recommendation for large-scale data models to meet the needs of the business solution is not aligned with the case study's details or contains factual or logical errors.	The recommendation for large-scale data models to meet the needs of the business solution is aligned with the case study's details and contains no factual or logical errors.

D3:DATA ANALYTICS SOLUTION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A recommendation for a large-scale data analytics solution that integrates the databases and processing systems identified in D1 not provided.	The recommendation for a large-scale data analytics solution that integrates the databases and processing systems identified in D1 contains factual or logical errors, is not aligned with the case study details or does provide querying, visualization, or reporting functionality.	The recommendation for a large-scale data analytics solution that integrates the databases and processing systems identified in D1 contains no factual or logical errors, is aligned with the case study details and provides querying, visualization, and reporting functionality.

D4:DATA ARCHITECTURE PLAN

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A description of required event orchestration, security, or other ancillary services required as part of the proposed analytics solution is not provided.	The description provided fails to identify required event orchestration, security, or other ancillary services required as part of the proposed analytics solution or contains factual or logical errors.	A description of required event orchestration, security, or other ancillary services required as part of the proposed analytics solution is provided and contains no factual or logical errors.

E1:ARCHITECTURAL AND TECHNOLOGICAL CHOICES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
An explanation for the reasoning behind *each* architectural and technological choice made during the design phase is not provided.	The explanation does not provide the reasoning behind *each* architectural choice or *each* technological choice, or the ex-	The explanation for the reasoning behind *each* architectural and technological choice made during the design phase is provided and contains no factual or logical errors.

planation contains factual or logical errors.

E2:ALIGNMENT WITH ORGANIZATIONAL NEEDS

NOT EVIDENT

An explanation for how the architectural and technological choices align with the business problem and organizational needs is not provided.

APPROACHING COMPETENCE

The response does not explain how the architectural choices align with the business problem or how the technological choices align with the business problem, or the response contains factual or logical errors.

COMPETENT

The response explains how the architectural and technological choices align with the business problem and organizational needs and contains no factual or logical errors.

F:FUTURE ENHANCEMENTS

NOT EVIDENT

A description of potential future enhancements or iterations to further improve the big data architecture is not provided.

APPROACHING COMPETENCE

The description of potential future enhancements or iterations to further improve the big data architecture is not aligned with the case study's details or contains factual or logical errors.

COMPETENT

The description of potential future enhancements or iterations to further improve the big data architecture is aligned with the case study's details and contains no factual or logical errors.

G:STEPS FOR IMPLEMENTATION

NOT EVIDENT

An overview of the proposed steps for implementing the designed solution is not provided.

APPROACHING COMPETENCE

The overview of the proposed steps for implementing the designed solution contains factual or logical errors or lacks specific details.

COMPETENT

The overview of the proposed steps for implementing the designed solution provides adequate details and contains no factual or logical errors.

H:APA SOURCES

NOT EVIDENT

The submission does not include in-text citations and references according to APA style for con-

APPROACHING COMPETENCE

The submission includes in-text citations and references for content that is quoted, paraphrased,

COMPETENT

The submission includes in-text citations and references for content that is quoted, paraphrased, or summarized and demonstrates

tent that is quoted, paraphrased, or summarized.	or summarized but does not demonstrate a consistent application of APA style.	a consistent application of APA style.

I:PROFESSIONAL COMMUNICATION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, negatively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.	This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.	This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote accurate interpretation and understanding.